

KINO D4 / CARRIER A0

Component—level draft + initial placement. No routed copper. NOT FOR FABRICATION.

Battery / NTC / USB coexistence / regulator compensation / final footprints and mechanical fit remain release gates.

P4 header and isolated control bus

File: 01_p4.kicad_sch

Camera 1 socket, power and signal isolation

File: 02_camera1.kicad_sch

Camera 2 socket, power and signal isolation

File: 03_camera2.kicad_sch

Camera 3 socket, power and signal isolation

File: 04_camera3.kicad_sch

Camera 4 socket, power and signal isolation

File: 05_camera4.kicad_sch

Camera enables and GPIO expansion

File: 06_controls.kicad_sch

Four camera currents and battery fuel gauge

File: 07_monitoring.kicad_sch

Motion, clock and board temperature

File: 08_sensors.kicad_sch

Cover, shutter and haptic interfaces

File: 09_controls.kicad_sch

USB—C PD sink and protected input

File: 10_usb_pd.kicad_sch

1S charger and battery entry

File: 11_charger.kicad_sch

5V boost and main load disconnect

File: 12_boost.kicad_sch

Hardware on/off and user indication

File: 13_power_button.kicad_sch

ERC supply declarations: Intended connector/regulator sources: see each symbol source note. Not a power—validation result.

GND

PACK_FUSED

MB_3V3

USB_VBUS

CHG_VBUS

SYS_5V

RTC_BACKUP

CAM1_SHUNT_OUT

CAM2_SHUNT_OUT

CAM3_SHUNT_OUT

CAM4_SHUNT_OUT

CAM1_3V3

CAM2_3V3

CAM3_3V3

CAM4_3V3

ENGINEERING DRAFT — NOT FOR FABRICATION

KINO

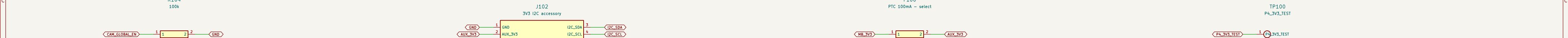
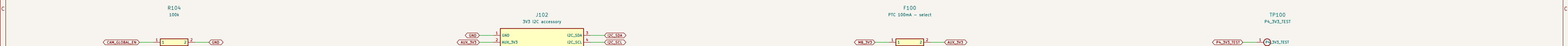
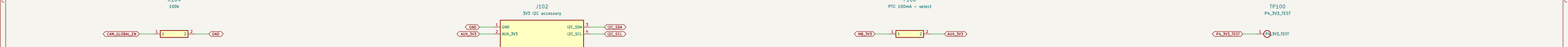
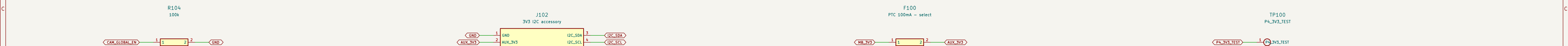
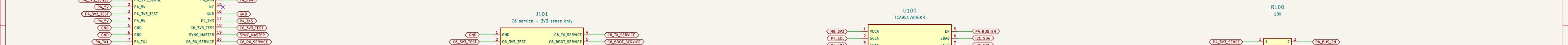
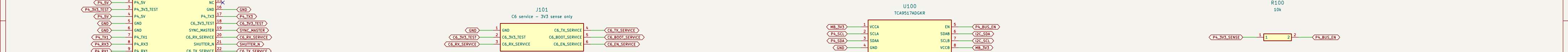
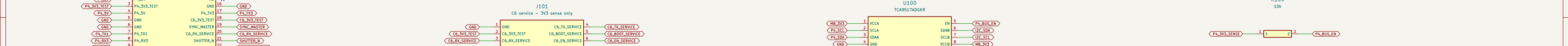
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Title: KINO D4 — four—XIAO carrier A0

Size: A2 Date: 2026—09—30
KiCad E.D.A. 10.0.6

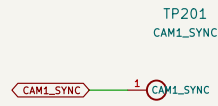
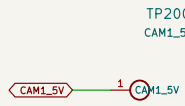
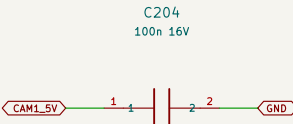
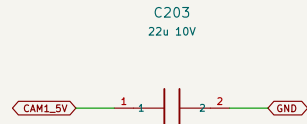
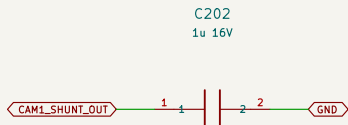
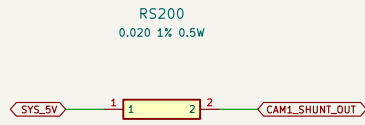
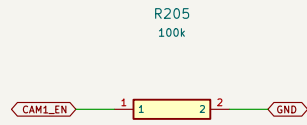
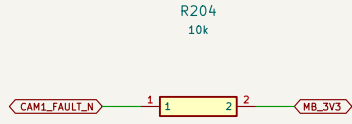
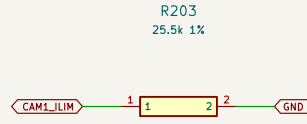
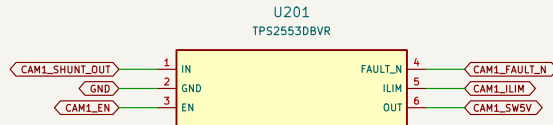
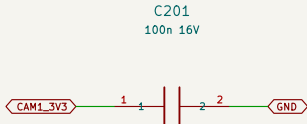
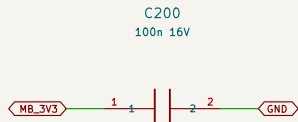
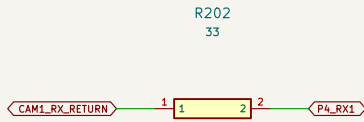
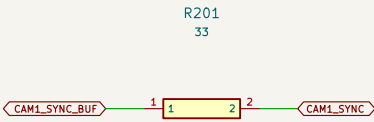
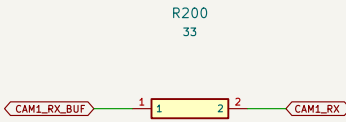
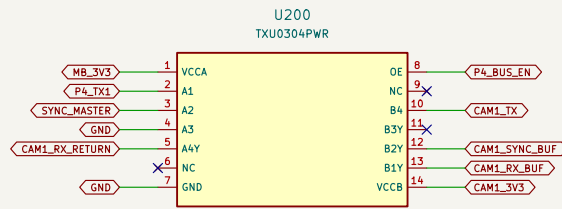
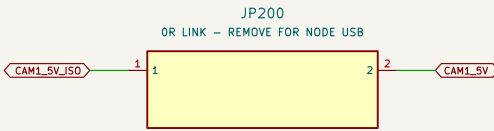
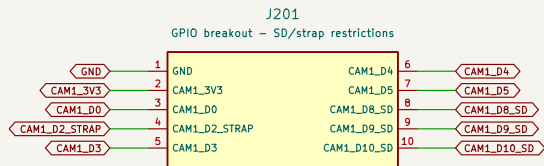
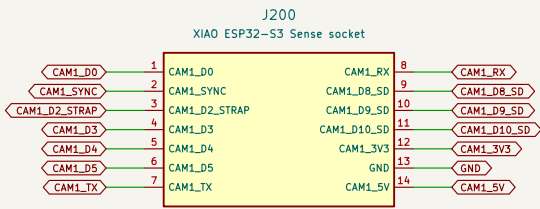
Rev: A0—DRAFT
Id: 1/14

P4 header and isolated control bus										
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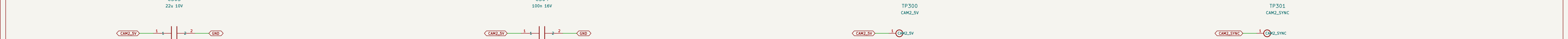
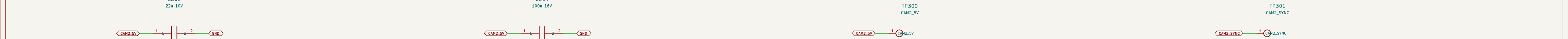
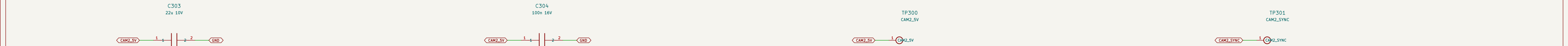
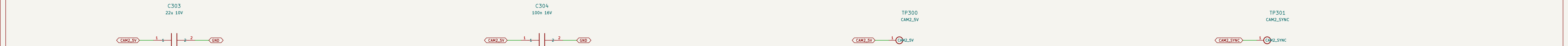
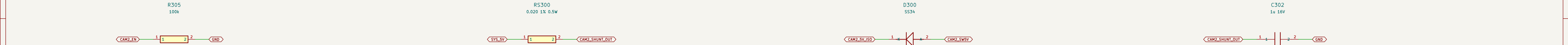
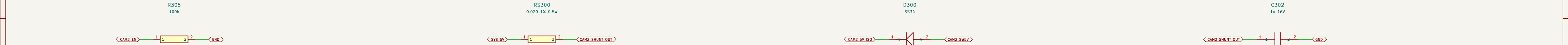
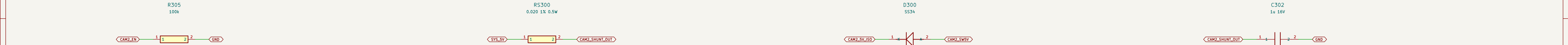
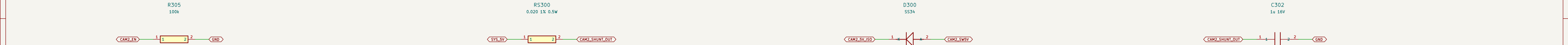
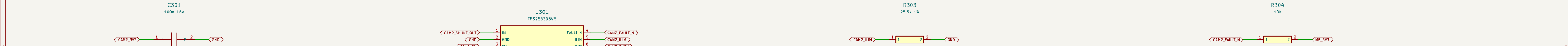
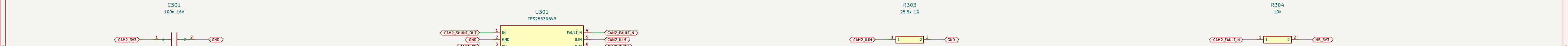
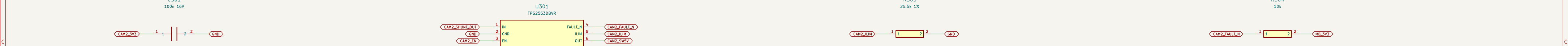
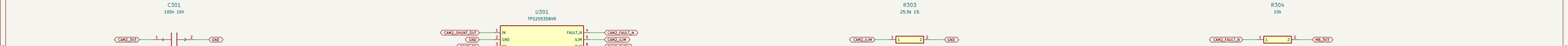
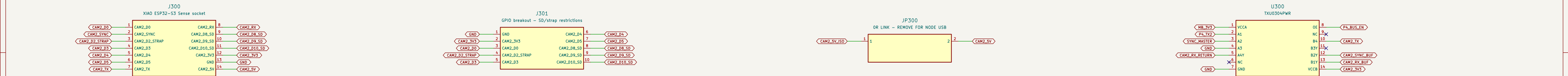
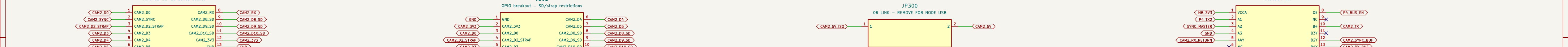
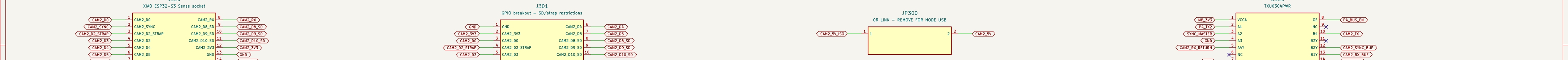
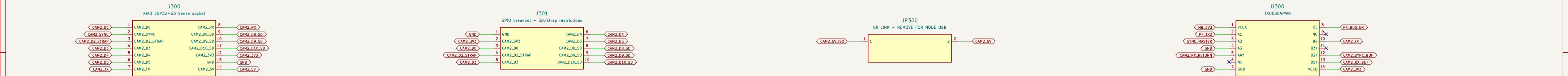
Camera 1 socket, power and signal isolation

Remove the camera power link (0 ohm, JP) BEFORE connecting node USB. Row centres 15.24 mm from official Seeed PCB/footprint. Optical datum remains to be verified.



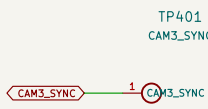
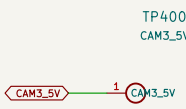
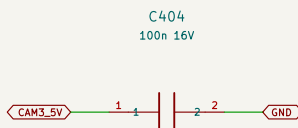
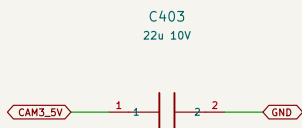
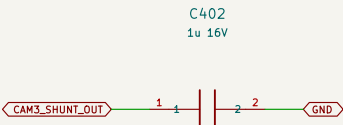
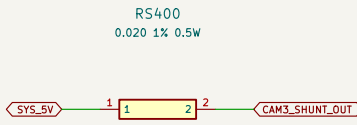
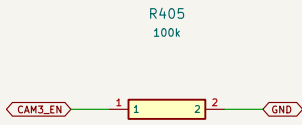
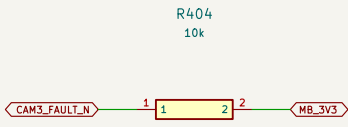
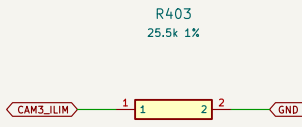
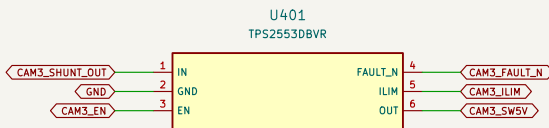
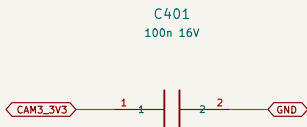
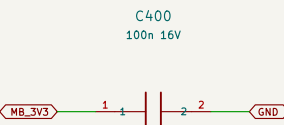
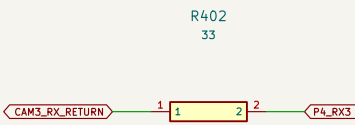
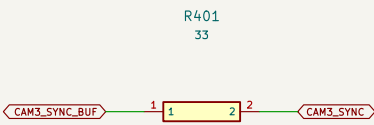
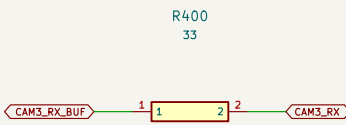
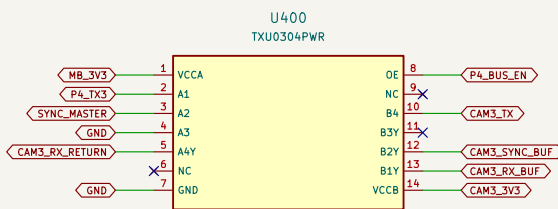
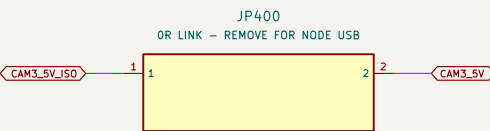
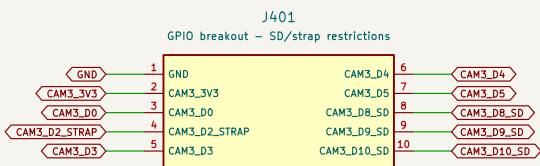
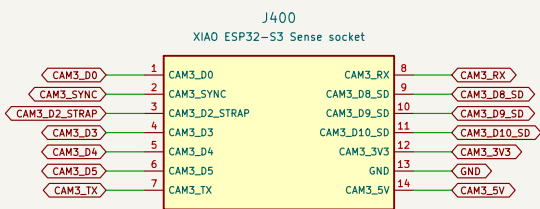
Camera 1 socket, power and signal isolation	2	3	4	5	6	7	8	9	10	11
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Remove the camera power link (0 ohm, JP) BEFORE connecting node USB. Row centres 15,24 mm from official Seeed PCB/footprint. Optical datum remains to be verified.



Camera 3 socket, power and signal isolation

Remove the camera power link (0 ohm, JP) BEFORE connecting node USB. Row centres 15.24 mm from official Seeed PCB/footprint. Optical datum remains to be verified.



ENGINEERING DRAFT - NOT FOR FABRICATION

KINO

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Title: Camera 3 socket, power and signal isolation

Size: A2 Date: 2026-09-30

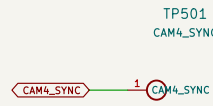
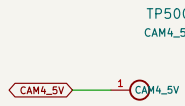
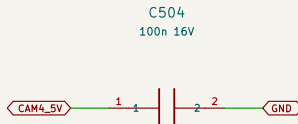
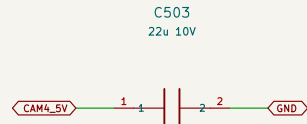
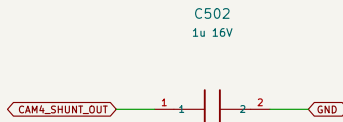
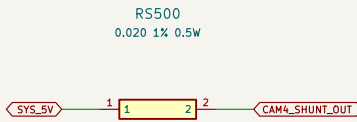
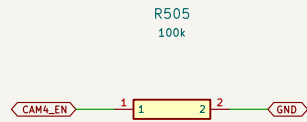
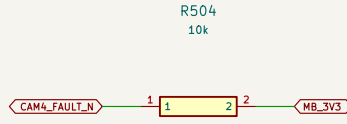
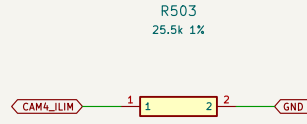
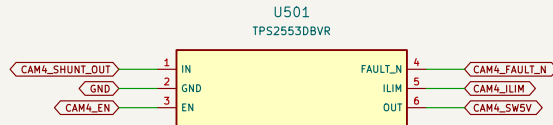
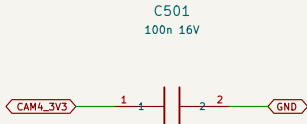
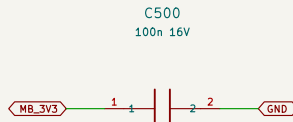
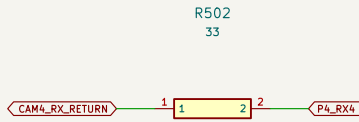
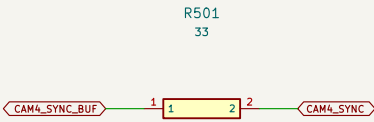
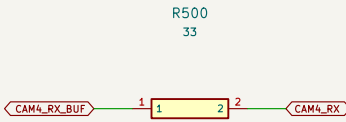
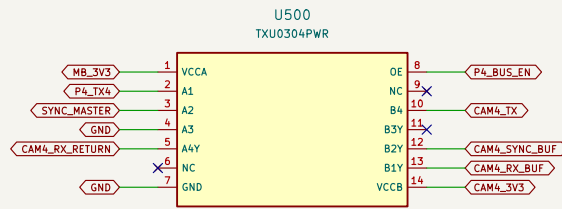
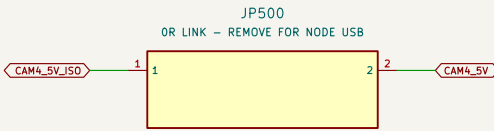
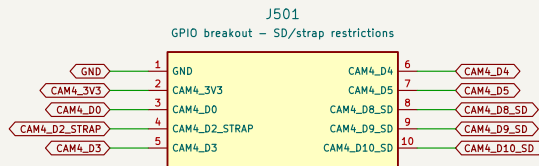
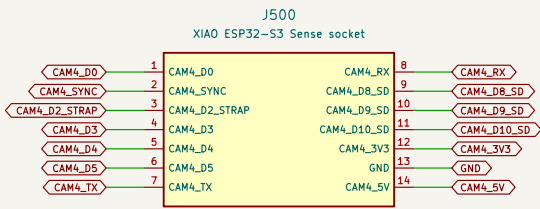
KiCad E.D.A. 10.0.6

Rev: A0-DRAFT

Id: 5/14

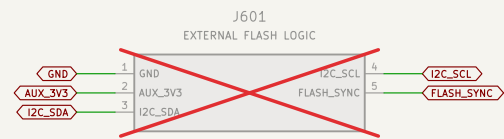
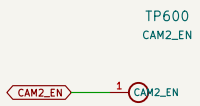
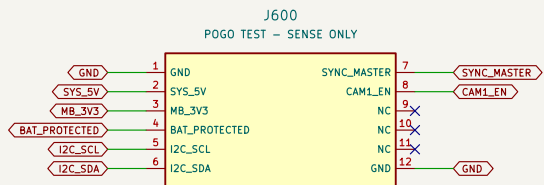
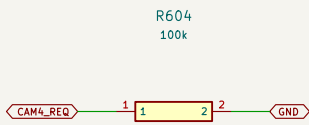
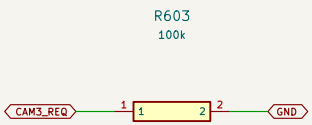
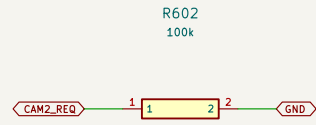
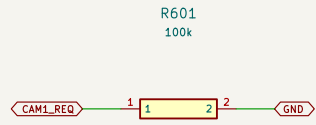
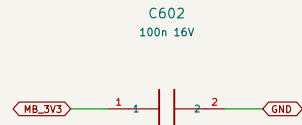
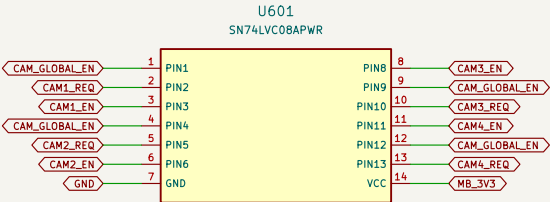
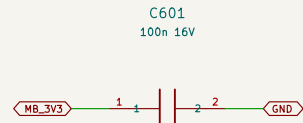
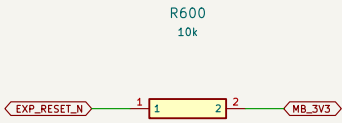
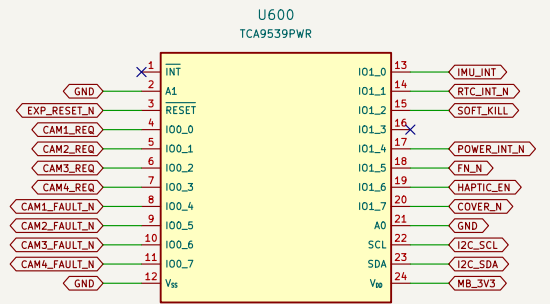
Camera 4 socket, power and signal isolation

Remove the camera power link (0 ohm, JP) BEFORE connecting node USB. Row centres 15.24 mm from official Seeed PCB/footprint. Optical datum remains to be verified.



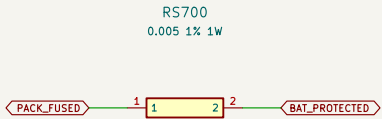
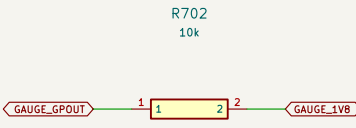
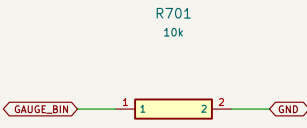
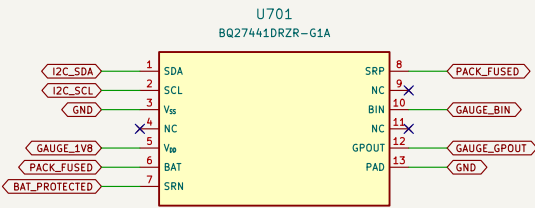
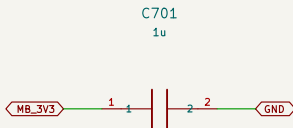
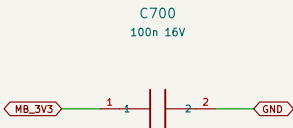
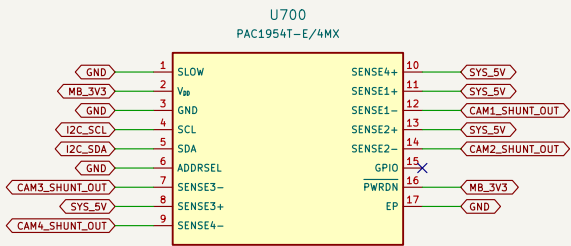
Camera enables and GPIO expansion

Camera outputs default off. Firmware must configure expander before enabling GPIO31. No spare P4 pin is consumed.



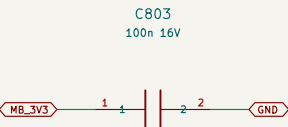
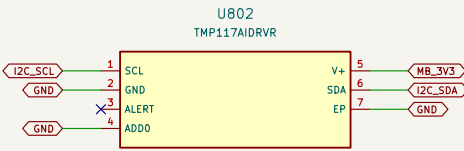
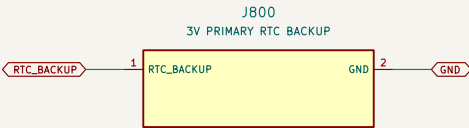
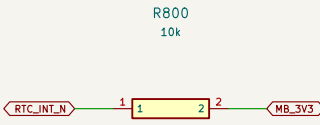
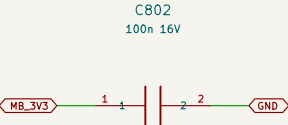
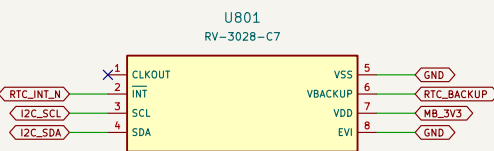
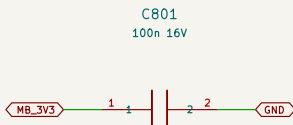
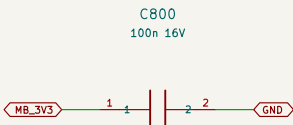
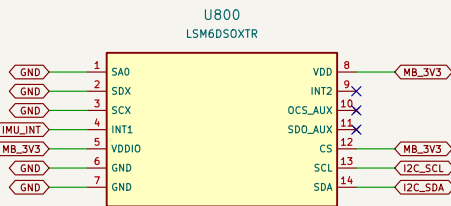
Four camera currents and battery fuel gauge

PAC1954 grounded ADDRSEL: 7-bit address 0x10; calibrate shunts. BQ27441 HIGH-SIDE 5 mOhm (T1 pins 7/8: SRP pack side, SRN system side). +/-25 mV range; 5 A = 25 mV. Gauge is factory--calibrated for 10 mOhm; firmware must write CC Gain for 5 mOhm.



Motion, clock and board temperature

IMU mode 1: axes marked on PCB. RTC backup charging must stay disabled for primary cell. No always-on wake promise.



ENGINEERING DRAFT – NOT FOR FABRICATION

KINO

Sheet: /Motion, clock and board temperature/

File: 08_sensors.kicad_sch

Title: Motion, clock and board temperature

Size: A2

Date: 2026-09-30

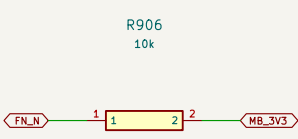
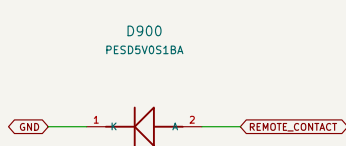
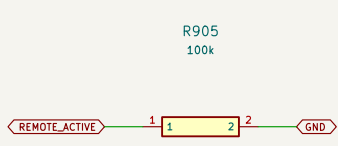
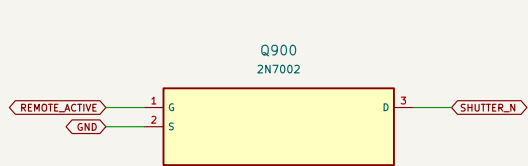
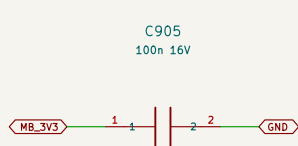
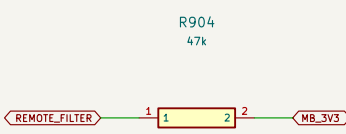
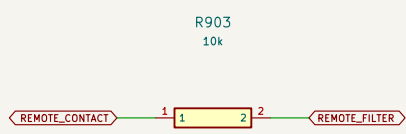
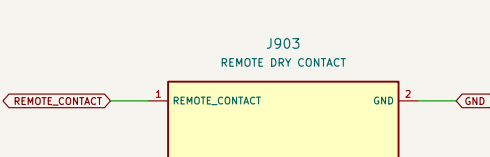
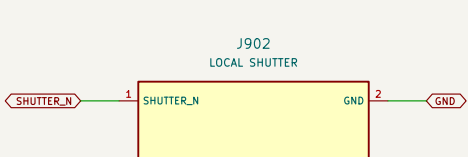
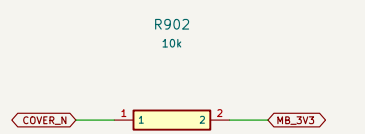
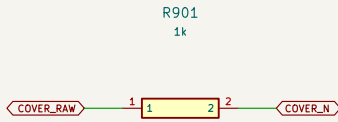
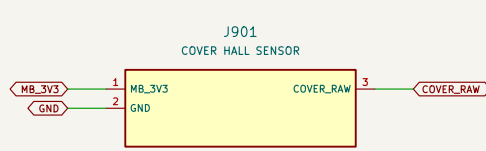
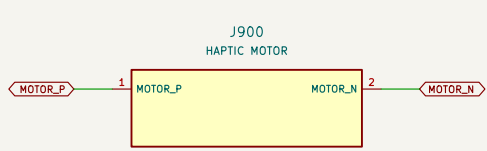
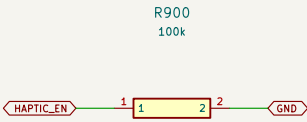
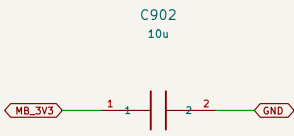
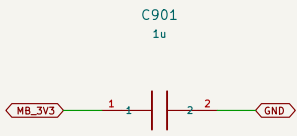
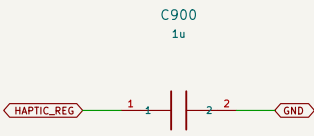
Rev: A0-DRAFT

KiCad E.D.A. 10.0.6

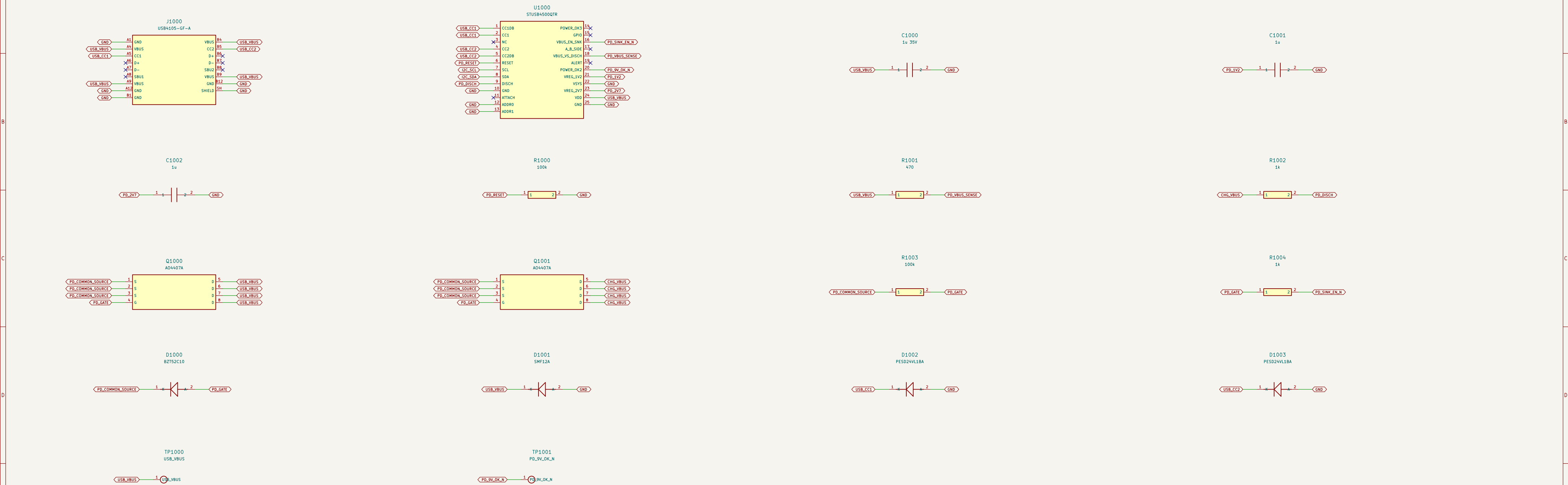
Id: 9/14

Cover, shutter and haptic interfaces

Passive dry-contact remote only. Haptic firmware must suppress vibration through exposure and settling. Motor MPN remains to be selected.

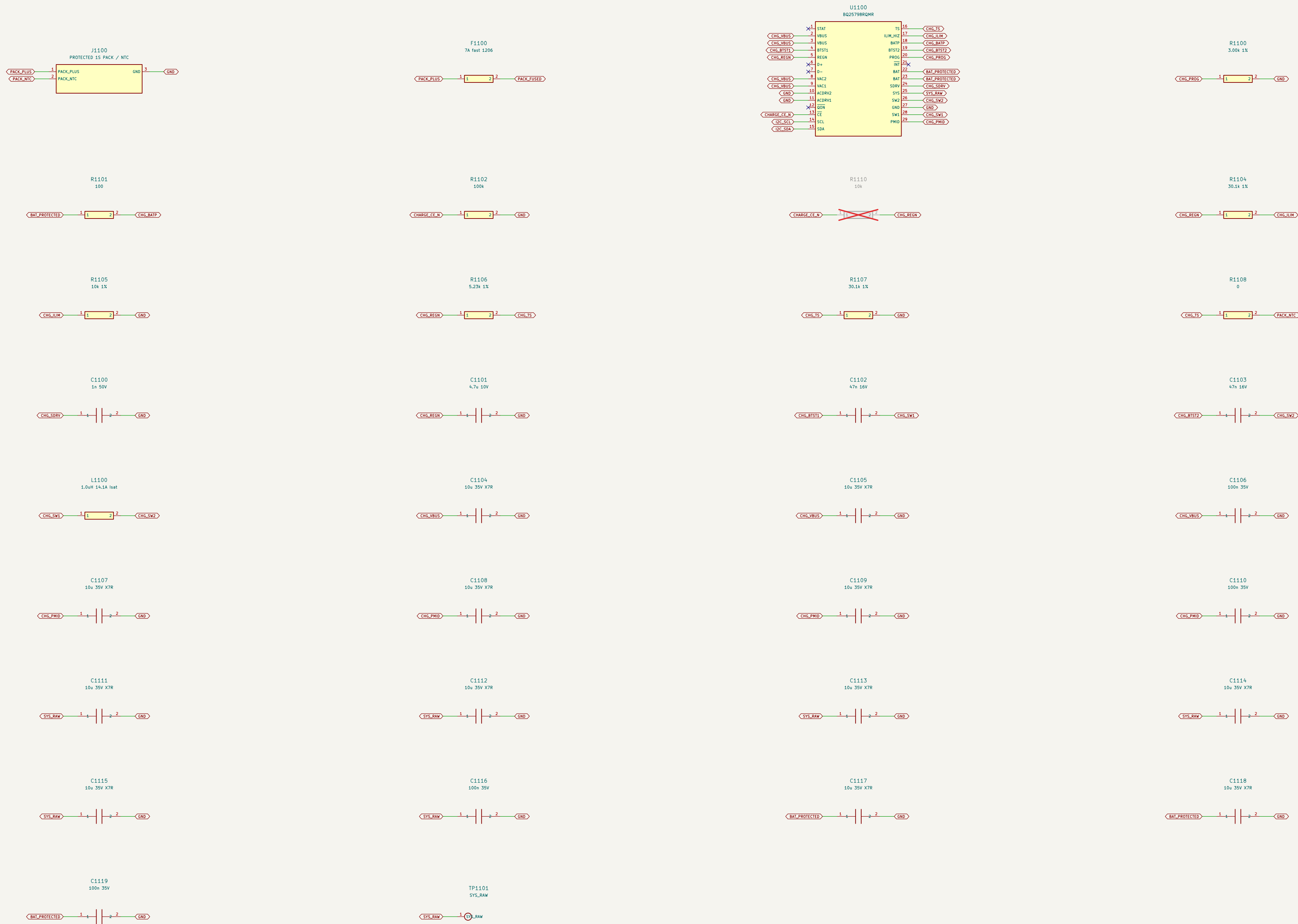


Charge-only port. STUSB4500 NVM must be programmed for 5V fallback and 9V/2A. Protection and gate transient review OPEN.



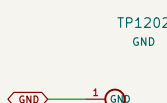
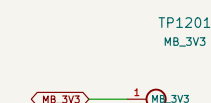
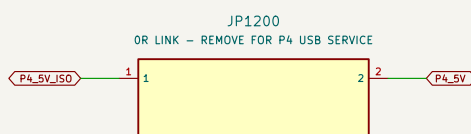
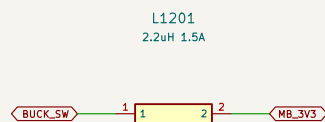
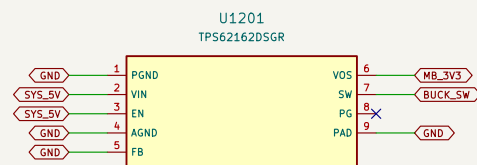
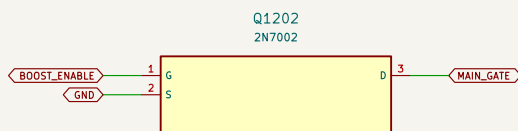
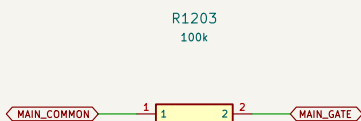
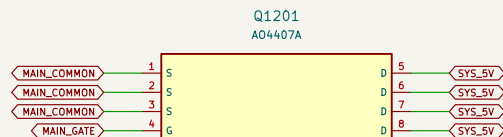
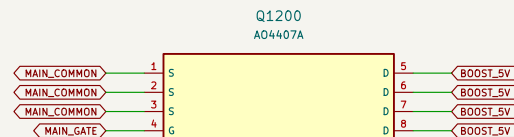
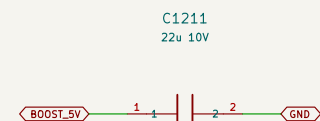
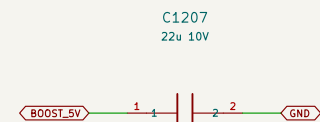
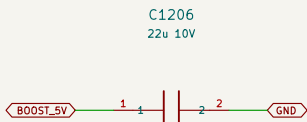
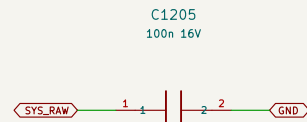
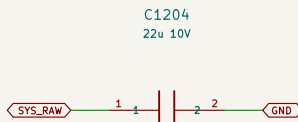
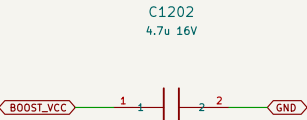
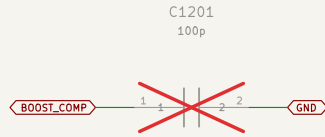
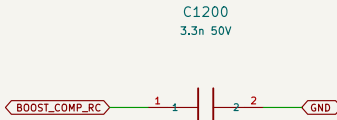
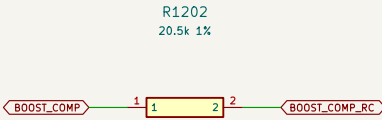
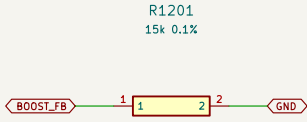
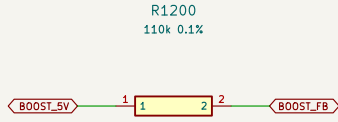
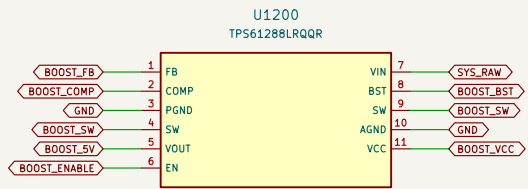
1S charger and battery entry

CE pulled LOW charges with the camera off. POR defaults 4.2 V / 1 A (PROG 3.00h, TI takes 7~1/7~2), 103M~2 pack NTC. TS window 1~60 C (TI RT1 5.2Nv / RT2 30.31s). Open or shorted NTC suspends charge. FR B1110 only to inhibit charging at bring-up. STAT (pin 1) is left open; the pin sits inside the charger power copper with no layout path out; firmware reads the charge state from the status registers over I2C.



5V boost and main load disconnect

5V = 0.6*(1+110k/15k). L Isat 19.6 A > ILIM 17.1 A max (TI 9.2.2.2). Comp per TI eq 12-14; fc 8-16 kHz, PM > 68 deg over Co 50-110 uF, L +/-20 %, Vin 3.0-4.3 V. Calculated: measure loop on bench.



ENGINEERING DRAFT - NOT FOR FABRICATION

KINO

Sheet: /5V boost and main load disconnect/

File: 12_boost.kicad_sch

Title: 5V boost and main load disconnect

Size: A2

Date: 2026-09-30

Rev: A0-DRAFT

KiCad E.D.A. 10.0.6

Id: 13/14

Hardware on/off and user indication

Power controller T58 pin map checked against manufacturer package drawing: EN is pin 7. KILL is pulled high until software requests shutdown. Long-hold target remains provisional. One green power LED (D1300, on MB_3V3, so lit only while the camera runs); there is no charge LED: the charger STAT pin has no layout path out, and charge state is shown by firmware from the charger status read over I2C.

